

Data Mining

Practical Session 2

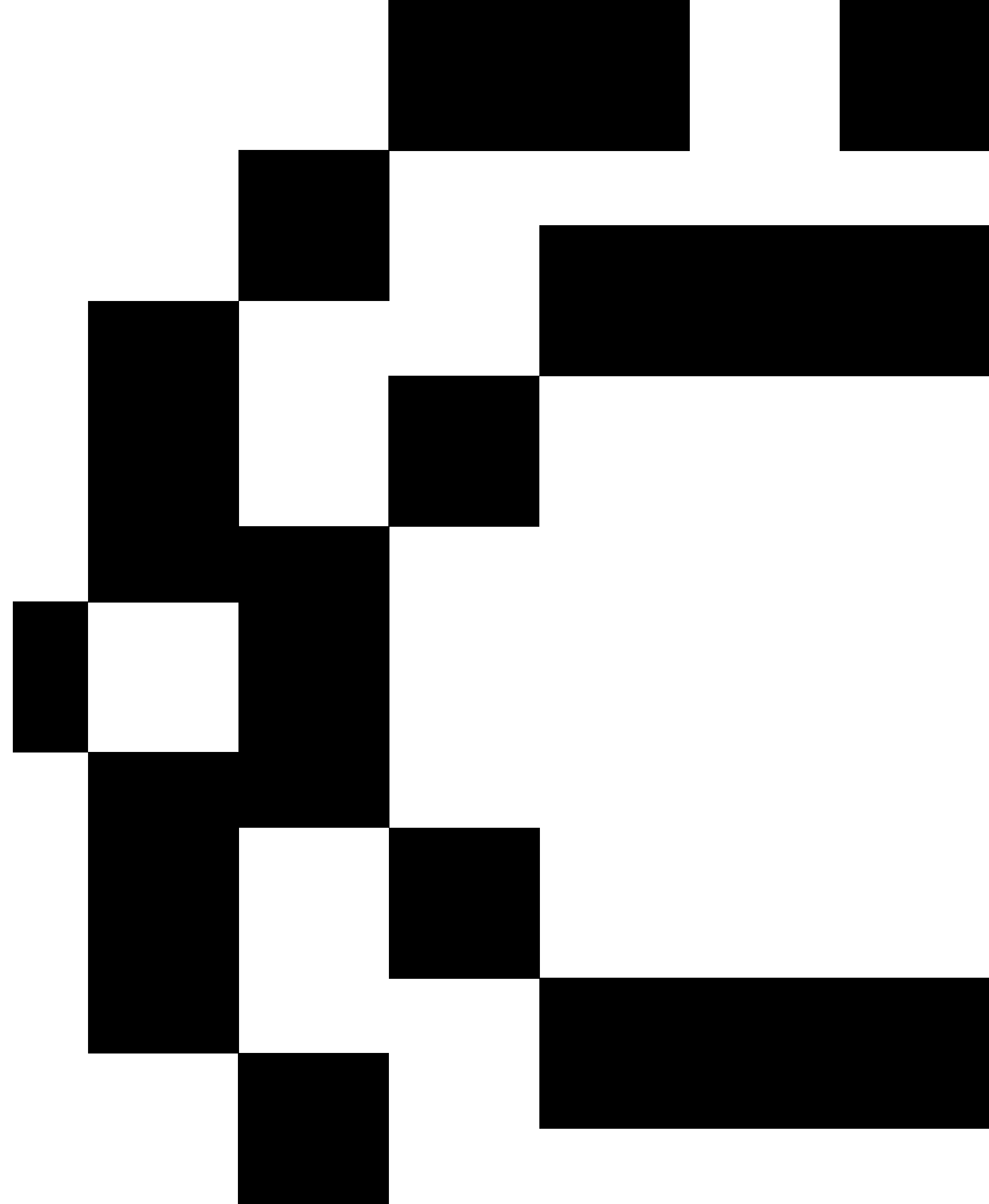
Fall Semester 2026-2027
Master's in Data Science and Advanced Analytics

Ana Pedro Caleiro

acaleiro@novaims.unl.pt

Yehor Malakhov

ymalakhov@novaims.unl.pt



Class Topics

- **Installing Anaconda**
- Creating a Data Mining environment
- Getting familiar with Jupyter notebooks and creating our first notebook
- Miniconda and using terminal



ANACONDA®

Open Source

Access the open-source software you need for projects in any field, from data visualization to robotics.

User-Friendly

With our intuitive platform, you can easily search and install packages and create, load, and switch between environments.

Trusted

Our securely hosted packages and artifacts are methodically tested and regularly updated.

<https://www.anaconda.com/download/success>



Products ▾

Solutions ▾

Resources ▾

Company ▾

Sign In

Get Demo >

Choose Your Download

Windows

Mac

Linux

Anaconda Distribution

Package and environment management with conda, Anaconda Navigator desktop app, and 600+ packages. Everything you need for data science.

GRAPHICAL INSTALLER

↓ [Windows 64-Bit Graphical Installer](#)

📘 Graphical install instructions

COMMAND LINE

📖 CLI install instructions

Miniconda

Lightweight installer that only includes conda, Python, and their dependencies. Install only what you need.

GRAPHICAL INSTALLER

↓ [Windows 64-Bit Graphical Installer](#)

📘 Graphical install instructions

COMMAND LINE

📖 CLI install instructions



Anaconda3-2025.12-2-Windows-x86_64.exe

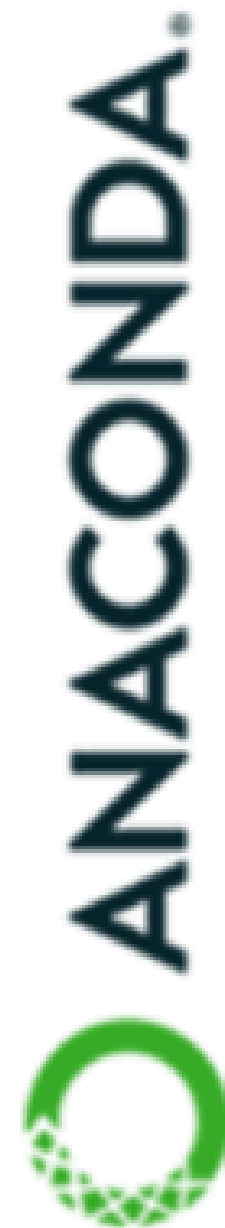
↓ 0.5/1.1 GB • 5 seconds left

Welcome to Anaconda3 2025.12-2 (64-bit) Setup

Setup will guide you through the installation of Anaconda3 2025.12-2 (64-bit).

It is recommended that you close all other applications before starting Setup. This will make it possible to update relevant system files without having to reboot your computer.

Click Next to continue.



Next...next...

Next >

Cancel

Class Topics

- Installing Anaconda
- **Creating a Data Mining environment**
- Getting familiar with Jupyter notebooks and creating our first notebook
- Miniconda and using terminal

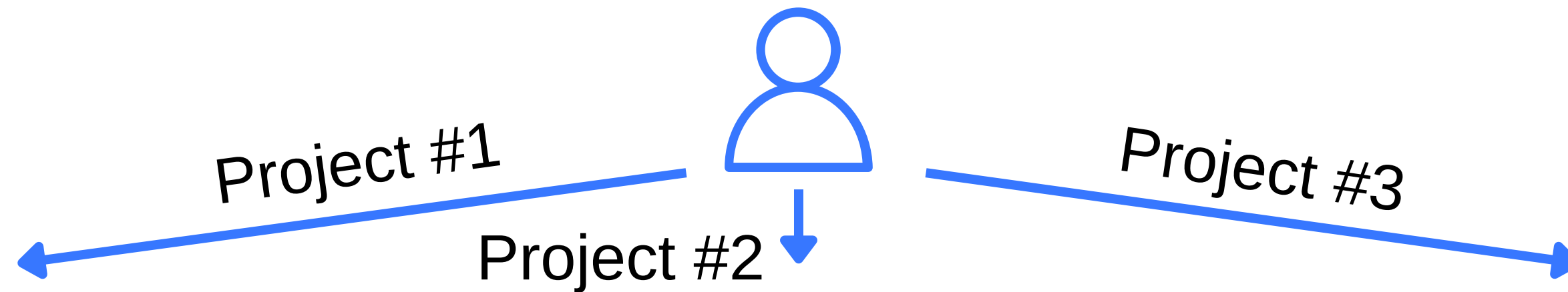
Virtual Environments

- Isolated spaces that contain per-project dependencies (a specific collection of installed conda packages)
- Using conda to manage environments:
 - Create, export, list, remove, and update environments
 - Switching or moving between environments (conda activate)
 - You can also share an environment file
- You can also use pip to manage environments



In other words...

You may need to use different versions of the same types of tools, or entirely different “environments”



Cooper (making a barrel)

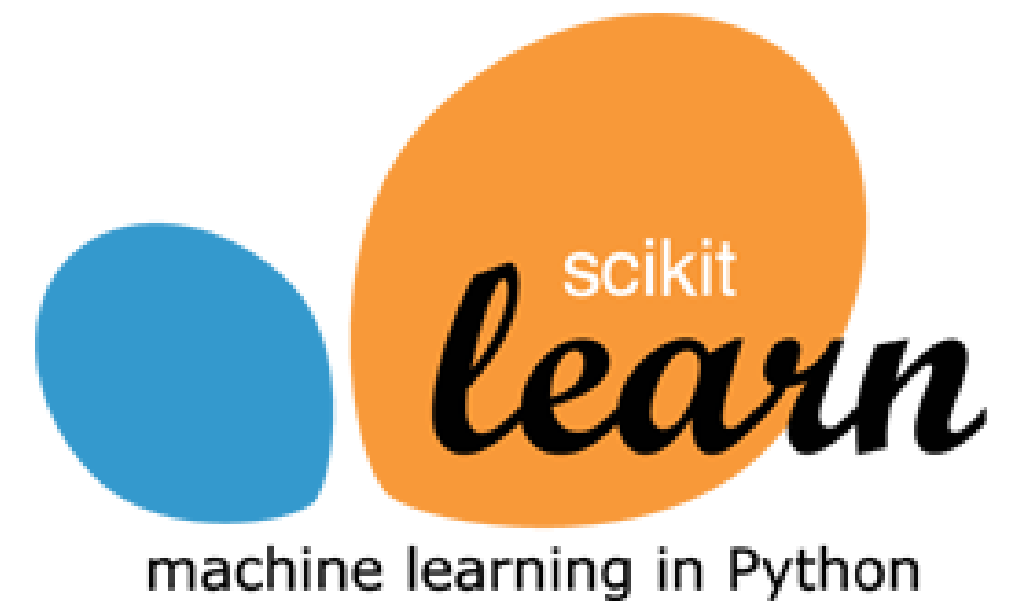


Formwork
(setting a structure)



Luthier
(making a guitar)

Python Packages



Creating new environment

File Help

 ANACONDA.NAVIGATOR

 Update Now

Connect 

 Home

 Environments

 Learning

 Community

Anaconda Quick Start Environments
Jump into pre-configured environments by project or industry. Clean dependencies, faster development.
Launch Your Environment

Documentation

Anaconda Blog



Search Environments



#1

base (root)



Installed

Channels

Update index...

Search Packages



Name



Description

Version



Create new environment



Name:

Fall2627

Location:

Packages:



Python

3.12.14



R

3.6.1

#3

Cancel

Create

#2



Create



Clone



Import



Backup



Remove

☒ anaconda-auth

 A client auth library for anaconda apis

 0.12.3

☒ anaconda-catalogs

 Client library to interface with anaconda cloud catalogs service

0.2.0

☒ anaconda-cli-base

 A base client endpoint supporting anaconda cli plugins

 0.7.0

Loading packages of C:\Users\egorm\anaconda3...

Cancel

Adding conda-forge channel

#2

Search Environments

Q

base (root)

Fall2627

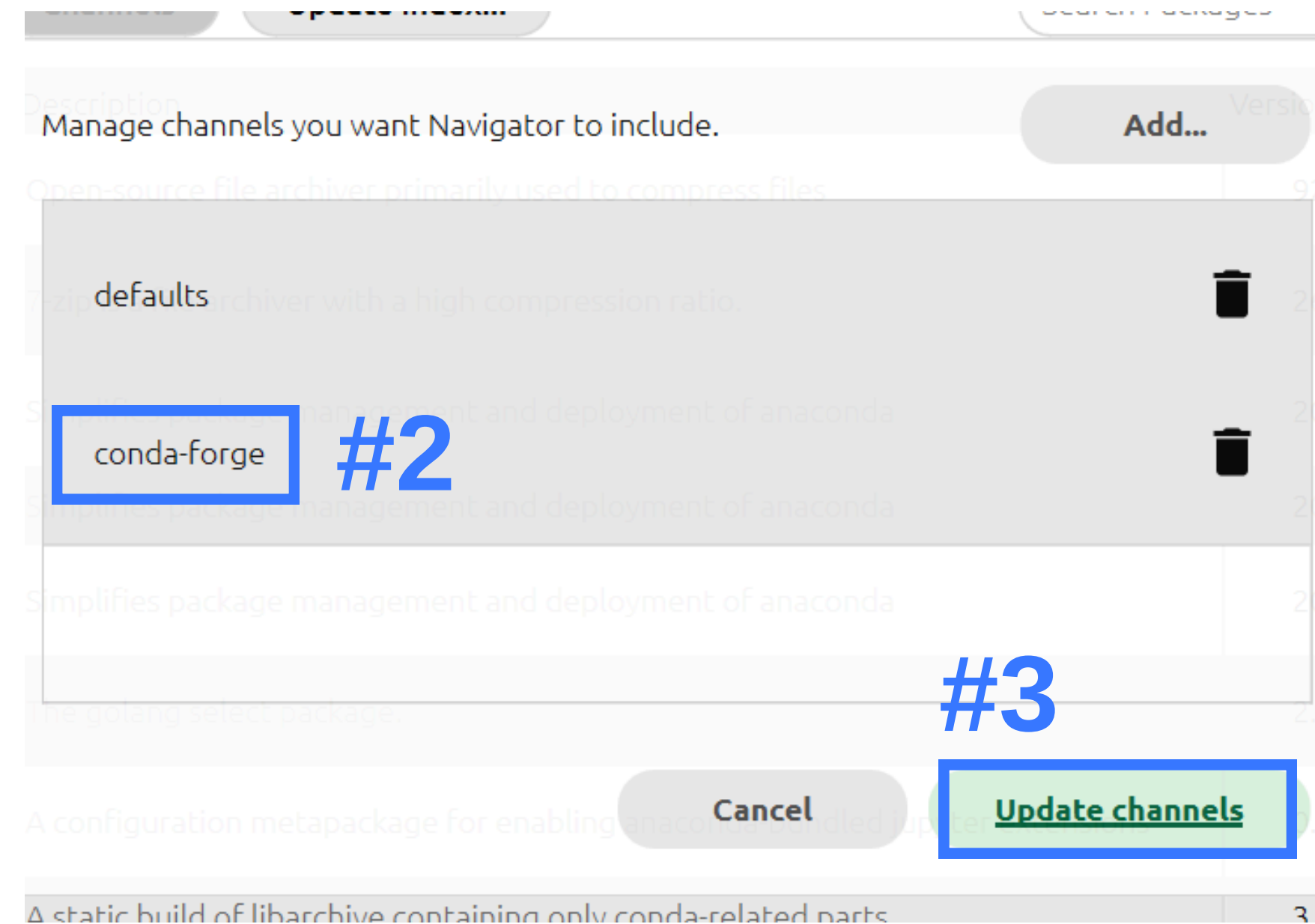
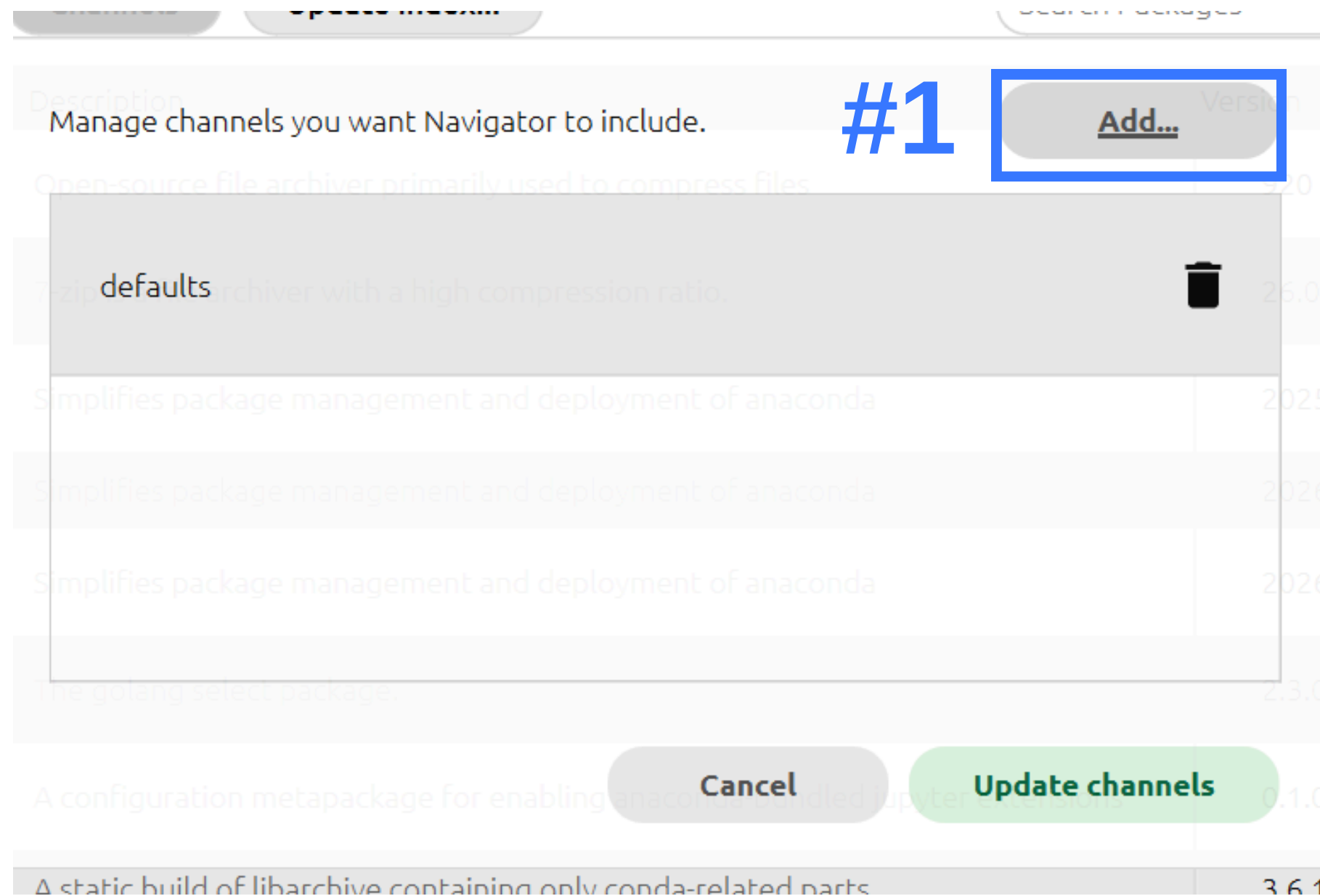
#1

Installed

Channels

Name	T	Description
✓ bzip2		High-quality d
✓ ca-certificates		Certificates fo
✓ libexpat		Expat xml par
✓ libffi		A portable for
✓ liblzma		A fast LZMA im

Adding conda-forge channel



Adding conda-forge channel and Finding libraries

All

Installed

Not installed

Updatable

Selected

✓ All

Channels

Update index... #1

	Description
	Open-source file archiver primarily used to
	7-zip is a file archiver with a high compress
☐ __anaconda_core_...	Simplifies package management and depl
☐ _anaconda_core	Simplifies package management and depl

Update index...

jupyter #3

	Version
uration metapackage for enabling anaconda-bundled jupyter extensions	0.1.0
	0.12.1
	0.2.0
	2.0.2
	3.4.8
	1.2.9

Installing libraries

Search package name, then select the check box on the left of the name

All

Channels

Update index...

#1 pandas

Name	T	Description	Version
☐ movingpandas			0.23.0
☐ narwhals		Extremely lightweight compatibility layer between pandas, polars, cudf, and modin	2.25.0
☐ nested-pandas			0.7.0
☐ pandarallel		An easy to use library to speed up computation (by parallelizing on multi cpus) with pandas.	1.6.5
☒ pandas #2		High-performance, easy-to-use data structures and data analysis tools.	3.0.5
☐ pandas-alive			0.2.4

- 1.jupyter
- 2.scikit-learn
- 3.scikit-image
- 4.numpy
- 5.pandas
- 6.ipynbwidgets
- 7.matplotlib
- 8.seaborn
- 9.minisom
- 10.ydata-profiling

After selecting, you can search the next library name, then select that, and so on

Installing libraries

after searching and selecting everything on the list,

you can clear the search bar by clicking the ‘x’

then filter the “Selected” option to see if you got everything

finally, click Apply

Selected

Channels

Update index...

Search Packages

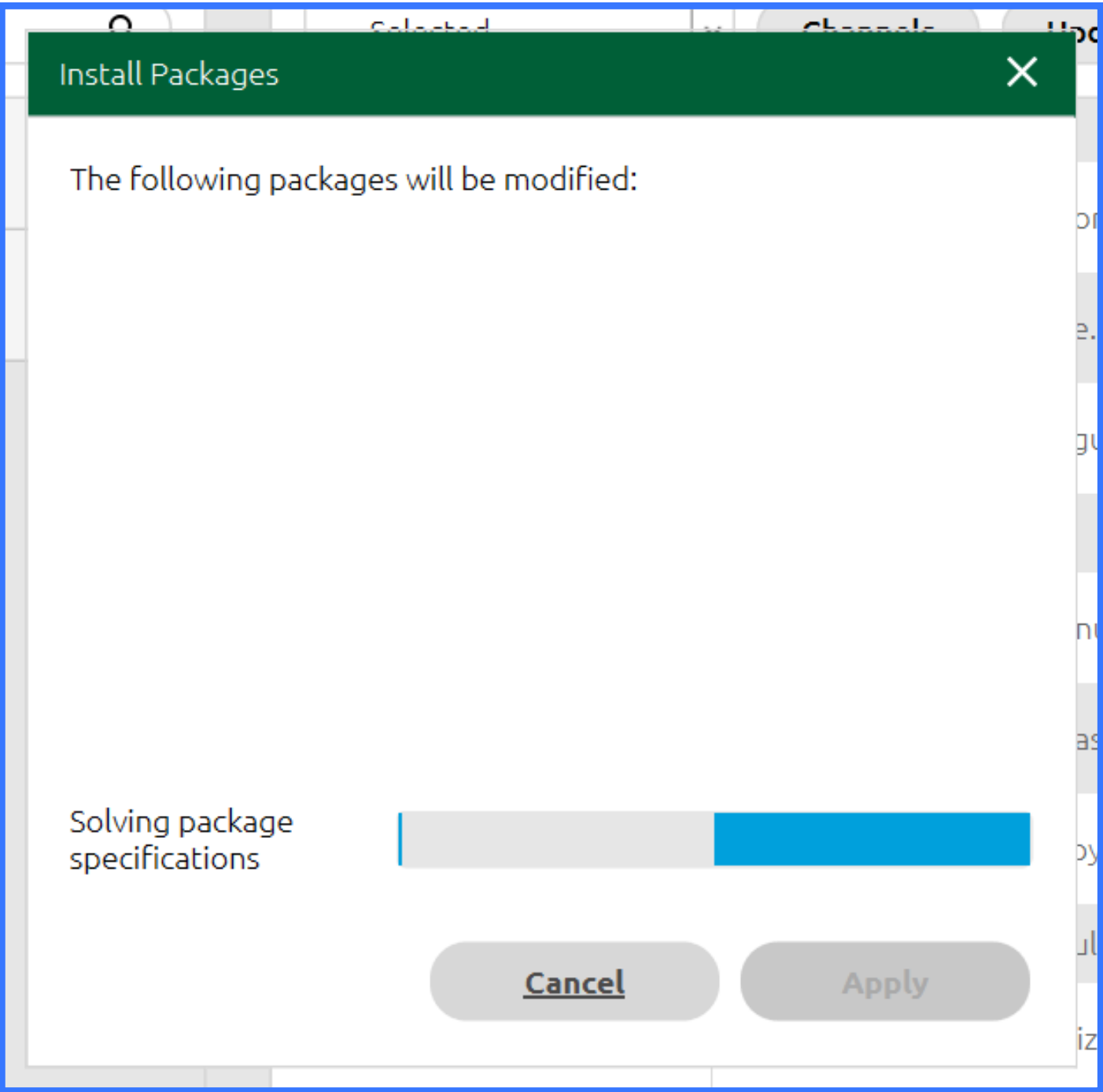
Name	T	Description	Version
☒ ipywidgets	☐	Interactive widgets for the jupyter notebook	8.1.9
☒ jupyter	☐	Jupyter metapackage. install all the jupyter components in one go.	1.1.1
☒ matplotlib	☐	Publication quality figures in python	3.11.1
☒ minisom	☐		2.3.6
☒ numpy	☐	Array processing for numbers, strings, records, and objects.	2.5.2
☒ pandas	☐	High-performance, easy-to-use data structures and data analysis tools.	3.0.5
☒ scikit-image	☐	Image processing in python.	0.26.0
☒ scikit-learn	☐	A set of python modules for machine learning and data mining	1.9.0
☒ seaborn	☐	Statistical data visualization	0.13.2
☒ ydata-profiling	☐	Generate profile report for pandas dataframe	4.18.4

10 packages available 10 packages selected

Apply

Clear

Installing libraries



Install Packages

1 package will be modified 274 packages will be installed

	Name	Unlink	Link	Channel	Action
124	*libxslt	-	1.1.43	conda-forge	Install
125	*libxml2-16	-	2.14.6	conda-forge	Install
126	*libxml2	-	2.14.6	conda-forge	Install
127	*libxcb	-	1.17.0	conda-forge	Install
128	*libwinpthread	-	12.0.0.r4.gg4f2fc60ca	conda-forge	Install
129	*libwebp-base	-	1.6.0	conda-forge	Install
130	*libtiff	-	4.7.2	pkgs/main	Install

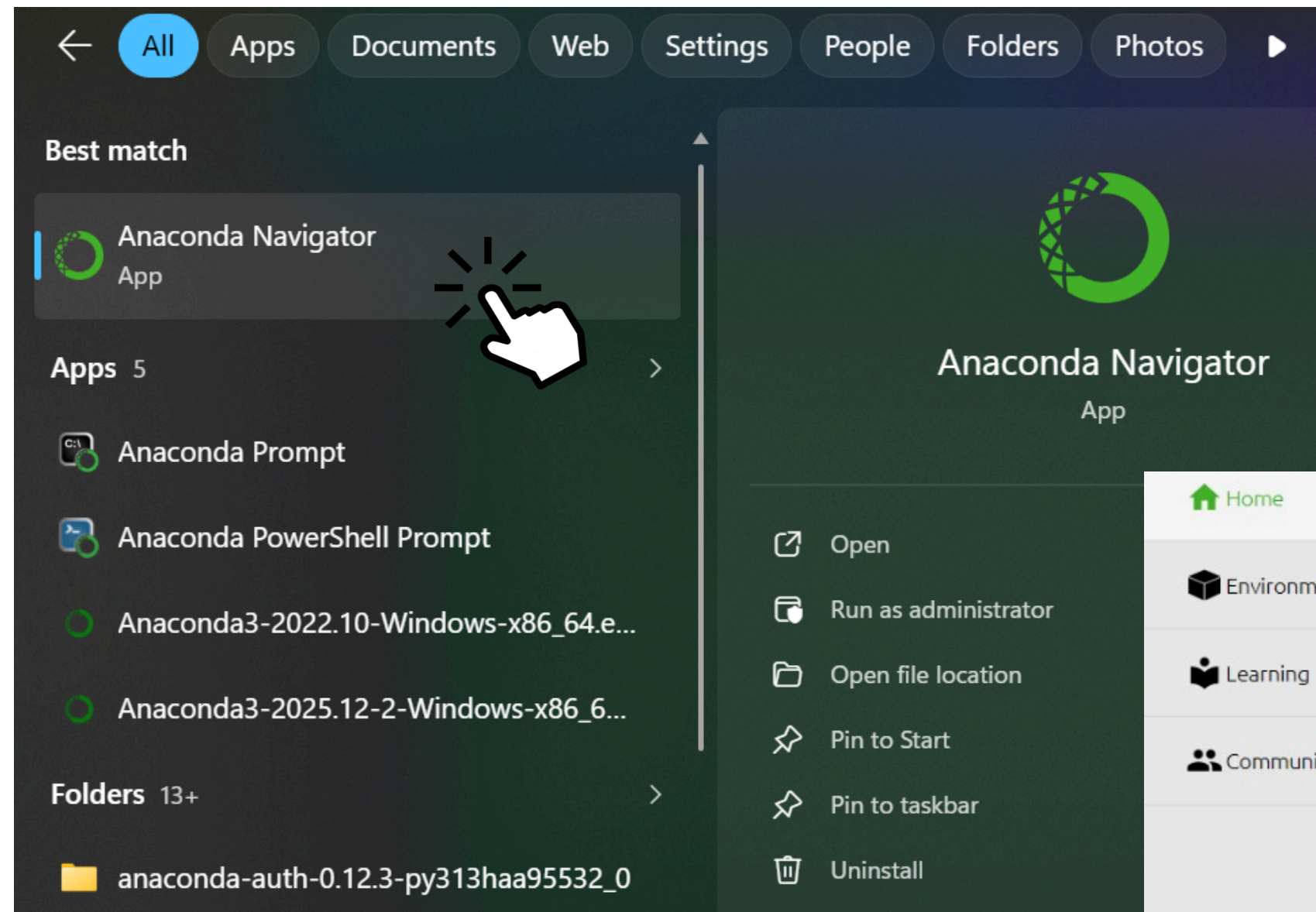
* indicates the package is a dependency of a selected packages

Cancel

Apply

Class Topics

- Installing Anaconda
- Creating a Data Mining environment
- **Getting familiar with Jupyter notebooks and creating our first notebook**
- Miniconda and using terminal

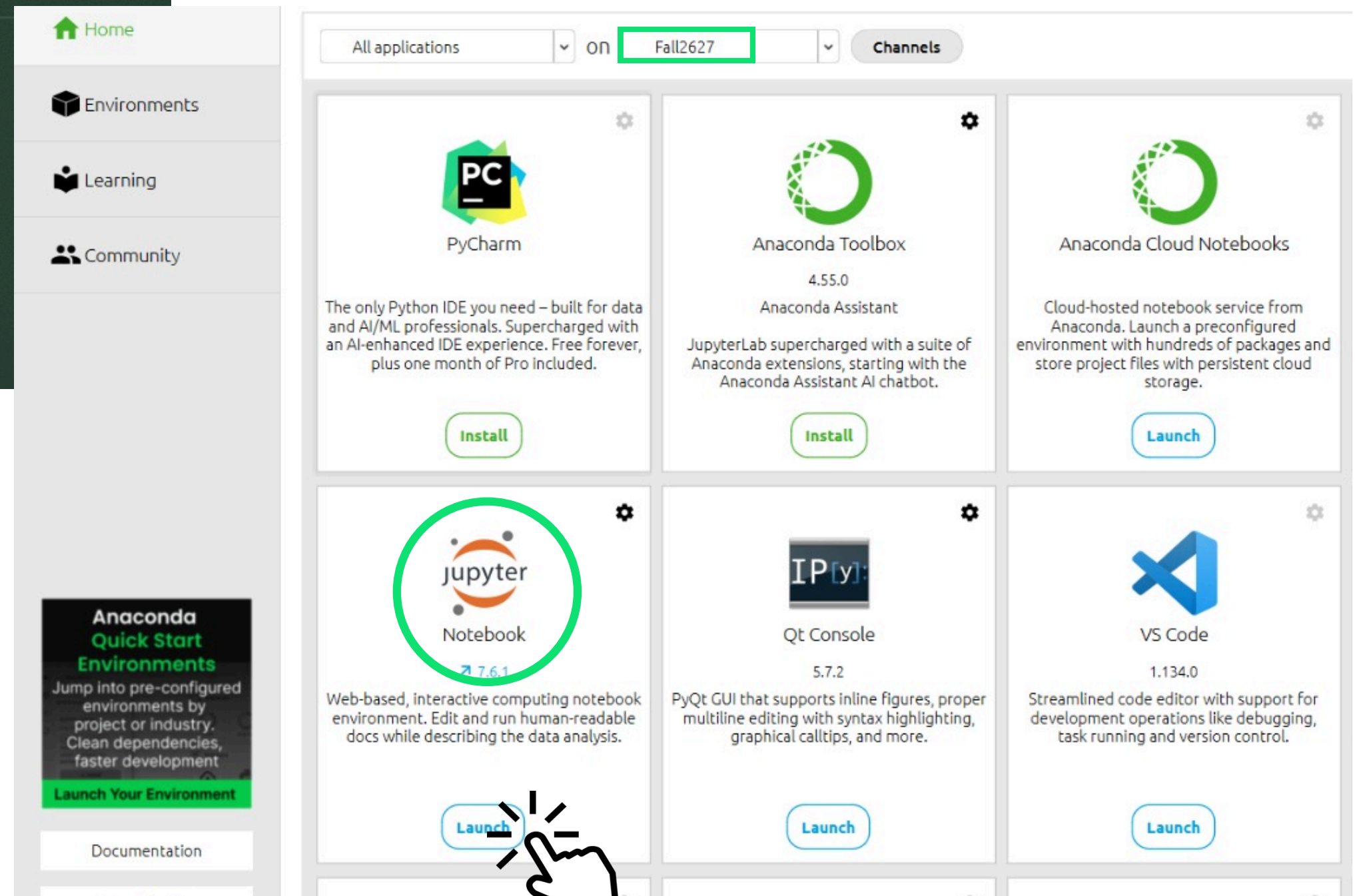


Search for
Anaconda Navigator

Change Environment
to **Fall2627**

Then find **Jupyter Notebook**

If it **doesn't appear** see
command line chapter



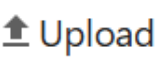
This is what you are supposed to see

 jupyter

File View Settings Help

Files Running

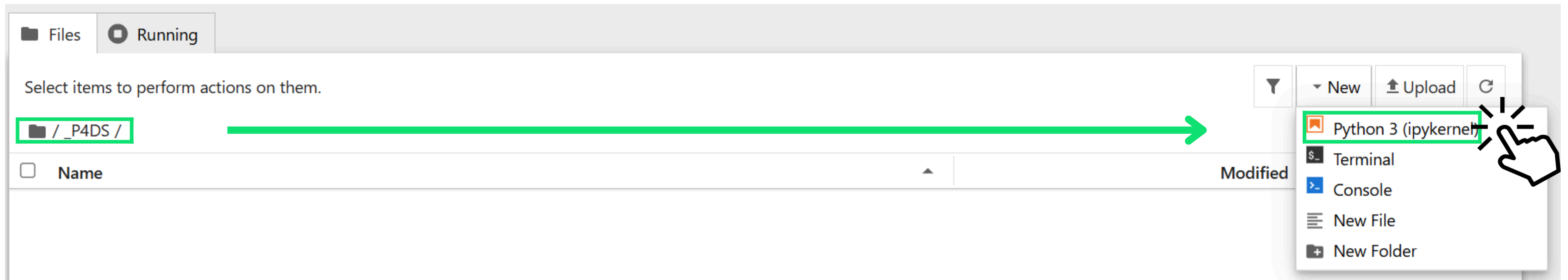
Select items to perform actions on them.

  New  Upload 

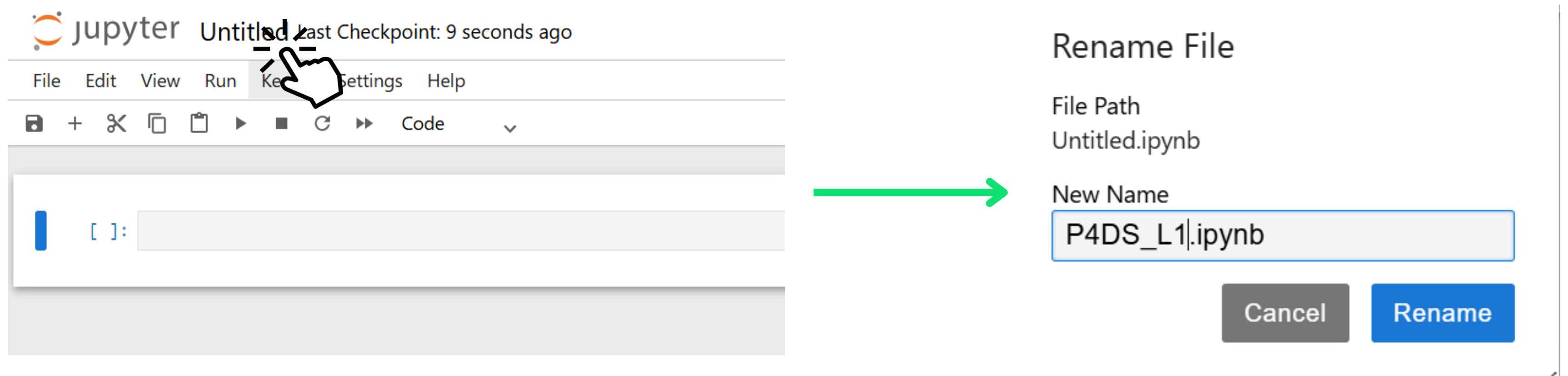
 /

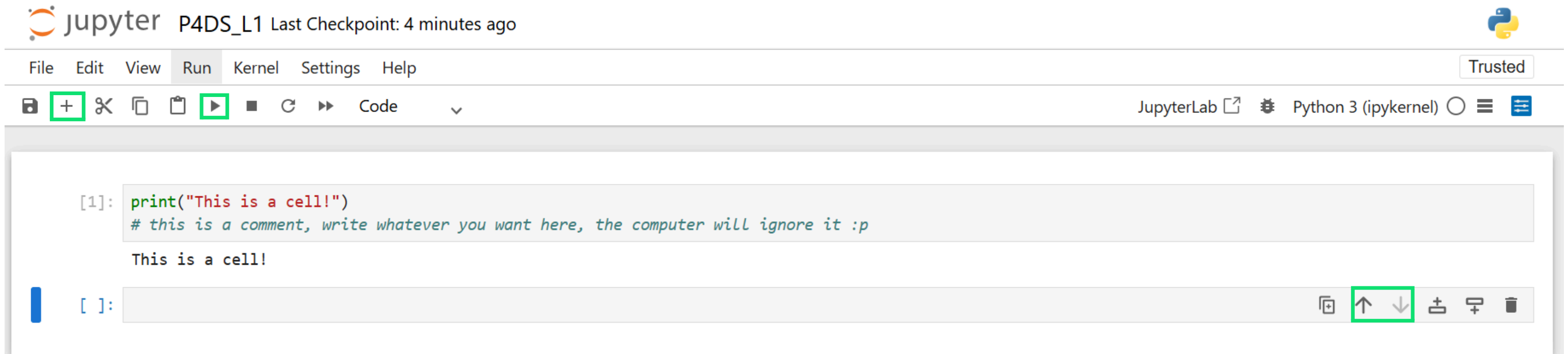
☐ Name	Modified	File Size
☐  _P4DS 	1 minute ago	
☐  anaconda	2 hours ago	
☐  classes_notebook	last month	
☐  Contacts	last year	
☐  Desktop	4 minutes ago	
☐  Documents	1 minute ago	
☐  Downloads	22 hours ago	
☐  Favorites	last year	
☐  gensim-data	last month	
☐  Links	last year	
☐  Microsoft	2 years ago	
☐  miniconda3	4 days ago	

You can choose a folder and then create a **new** notebook



You can also rename the notebook



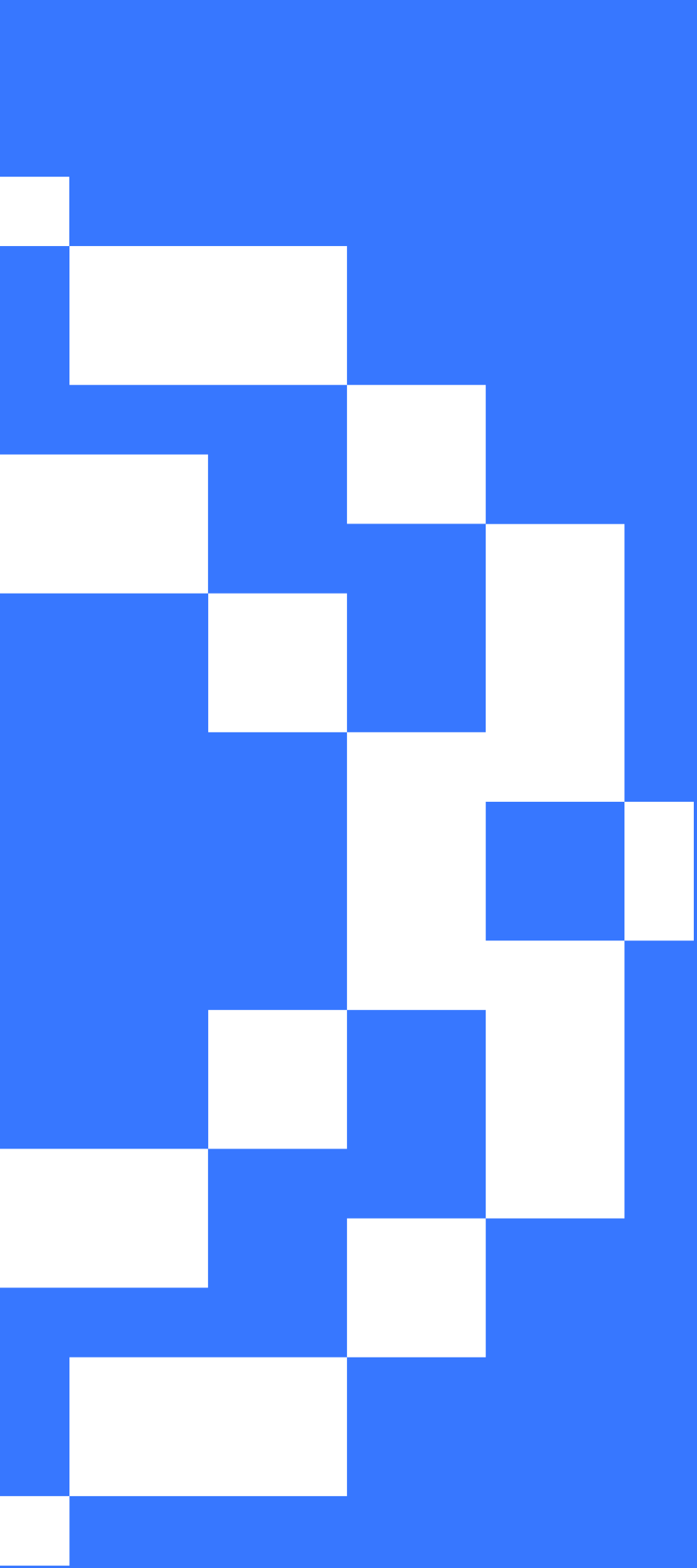


You can run a cell (“perform” / “activate” / “run” your code) by clicking the **Run Button (▶)**

Alternatively, you can run a cell by clicking **Shift + Enter**.

You can also create a new cell by clicking the “+” button.

Finally, you can reorder the cells by clicking the **arrows**.



Class Topics

- Installing Anaconda
- Creating a Data Mining environment
- Getting familiar with Jupyter notebooks and creating our first notebook
- **Miniconda and using terminal**

Miniconda

Miniconda is a lighter installer for **conda**, that doesn't contain **Anaconda Navigator**.

Thus, creation and management of environments is done through **terminal**.

To install you can either follow the same steps as for Anaconda Distribution or follow a shell installer guide:

- Windows
- macOS
- Linux

Miniconda

Lightweight installer that only includes conda, Python, and their dependencies. Install only what you need.

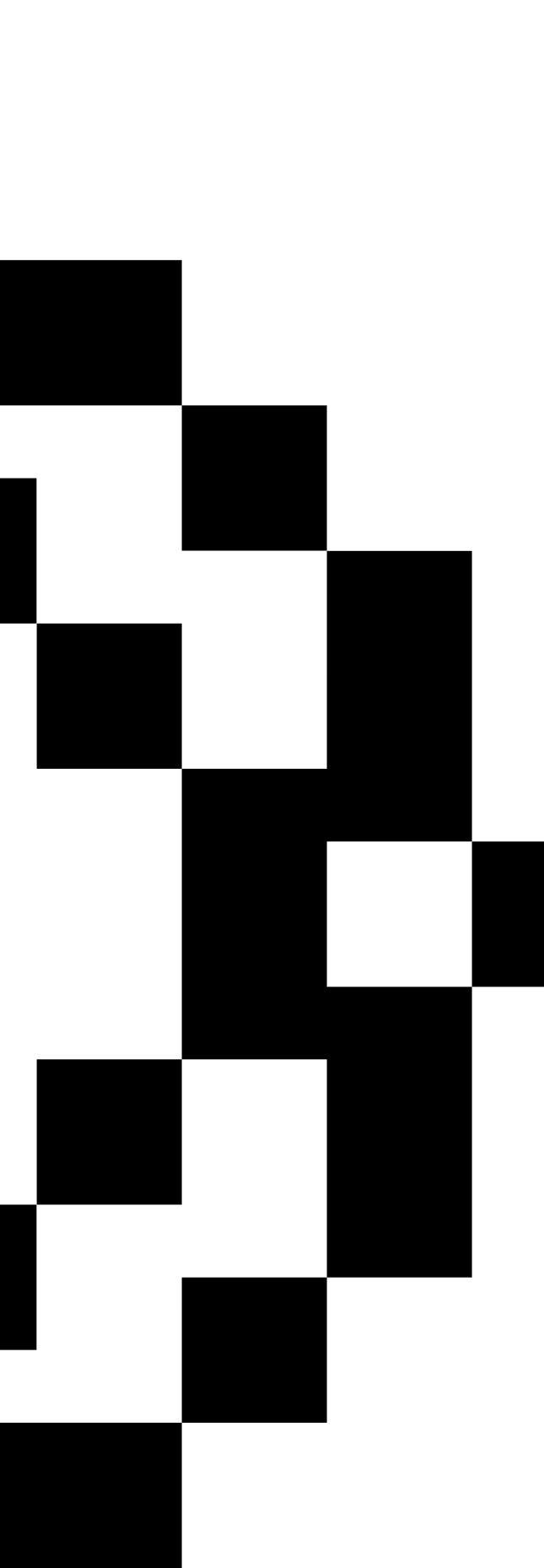
GRAPHICAL INSTALLER

[↓ Windows 64-Bit Graphical Installer](#)

[i Graphical install instructions](#)

COMMAND LINE

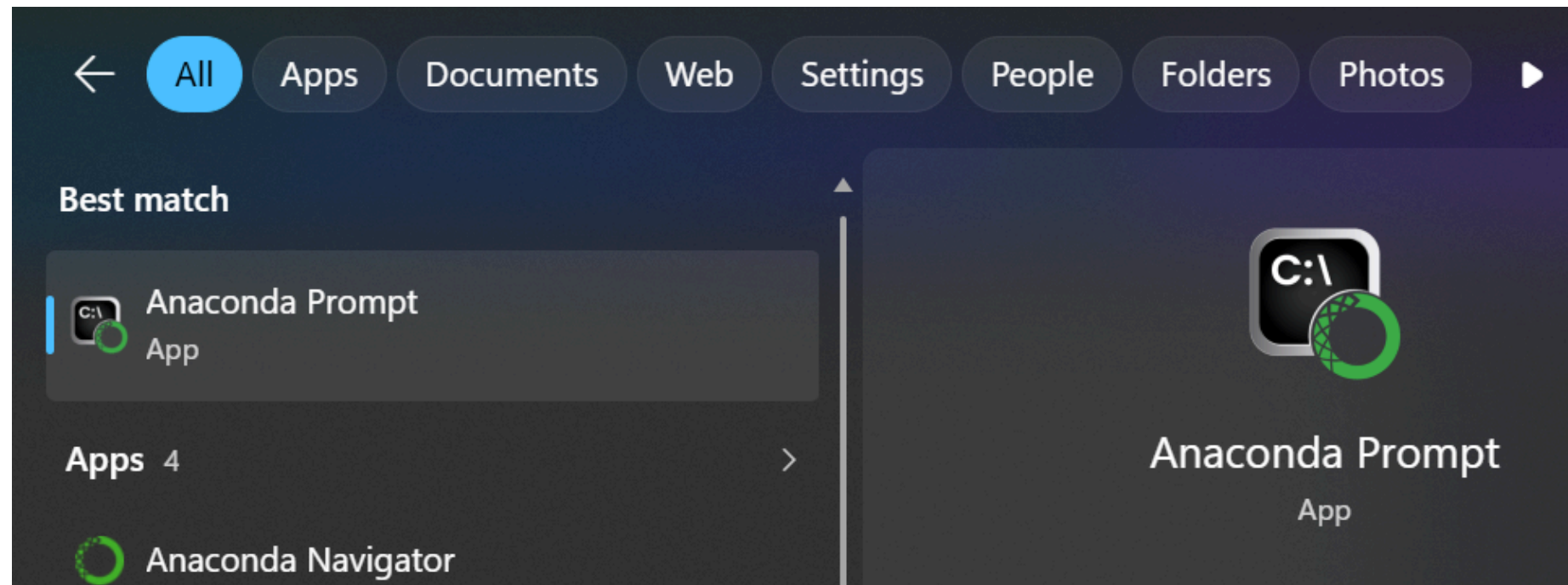
[> CLI install instructions](#)



Creating environment using the command line

- Open command line
 - **Windows:** Open Anaconda Prompt (miniconda3)
 - **Linux/Mac:** Open Terminal
- Type in following commands
 - `cd Downloads`
 - `conda env create -f fall2627_env.yml`

Note, that for this commands to work you should have the **fall2627_env.yml** file, available on GitHub, in your computer's **Downloads** folder



To enter jupyter start by opening **terminal** on [Linux/Mac](#) or **Anaconda Prompt** on [Windows](#)

```
Anaconda Prompt
(base) C:\Users\egorm>conda activate Fall2627
(Fall2627) C:\Users\egorm>
```

Change environment by writing
conda activate Fall2627

```
Anaconda Prompt
(base) C:\Users\egorm>conda activate Fall2627
(Fall2627) C:\Users\egorm>jupyter notebook
```

And type in
jupyter notebook

Useful Links

- Installing Anaconda
 - <https://www.anaconda.com/download/success>
 - Documentation: <https://www.anaconda.com/docs/main>
- Download notebooks and other materials:
 - [Moodle / Practical Sessions / Lab 02](#)
 - [Course GitHub](#)
- **Optional:** GitHub Desktop
 - <https://desktop.github.com/download>
 - Sign up for GitHub: <https://github.com/signup>

Obrigado!

Acreditações e Certificações da NOVA IMS



Cofinanciado por

